

# Industrial Component Anomaly Detection

## Project Presentation

Unsupervised defect classification and pixel-level anomaly localisation from normal-only manufacturing parts.

Unsupervised AD

Deep Transfer Learning

Zero-Leakage Protocol

### TEAM

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MVTEC AD 2D Dataset

Selected Examples · Objects & Textures

CATEGORY

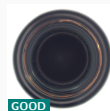
CONFORMING

DEFECTIVE

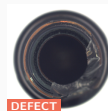
**Bottle**

Broken rim

GOOD



DEFECT



**Capsule**

Scratch

GOOD



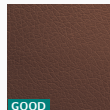
DEFECT



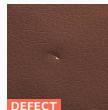
**Leather**

Surface cut

GOOD



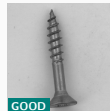
DEFECT



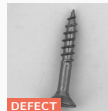
**Screw**

Thread defect

GOOD



DEFECT



# Data Exploration

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# The MVTec AD Dataset – Class Imbalance

## Dataset at a glance

- 15 manufacturing categories: 10 objects + 5 textures
- > 5000 images,  $\approx$  1200 anomalous images
- Normal-only training; test set has pixel-level anomaly ground truth

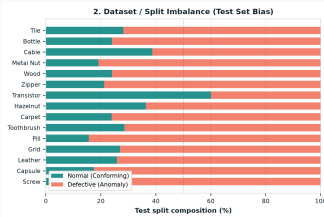
## Accuracy paradox

- Median anomalous-pixel area: **1.54%**
- Predicting *all-normal* yields >98.5% pixel accuracy while catching zero defects.

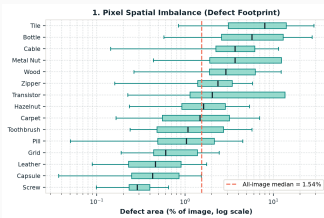
**Kruskal-Wallis:**  $H(14) \approx 626.5$ ,  $p < 0.001$

- $43\times$  spread in defect size across categories.

TEST-SET CLASS IMBALANCE (images)



PIXEL-AREA IMBALANCE (defect region %)



# Defect Taxonomy & Spatial Distribution

Two fundamentally different defect regimes:

## Structural Objects (e.g. Screw, Cable)

- Defects cluster at specific locations
- Strong positional bias (shortcut risk)
- Foreground/background separation critical

## Continuous Textures (e.g. Grid, Leather)

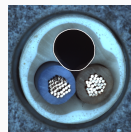
- Defects broadly distributed across surface
- Rich augmentation needed (rotation, scale)
- No foreground mask required

**Covariate shift:** Screw, Leather, Grid show measurable brightness/contrast drift between train and test normal images – motivates photometric augmentation.

### OBJECTS – LOCALISED DEFECTS (contour = defect boundary)



Screw thread

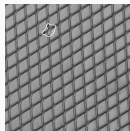


Cable missing

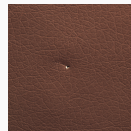


Capsule scratch

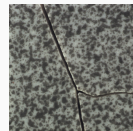
### TEXTURES – DISTRIBUTED DEFECTS (defect across full surface)



Grid glue strip



Leather cut



Tile crack

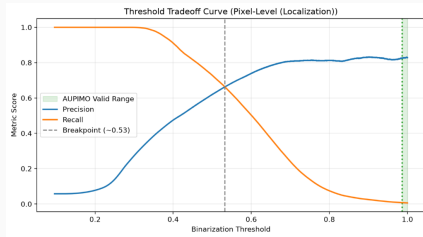
# Metric Selection: AUPIMO & F1-Score

## The Problem with Pixel-AUROC

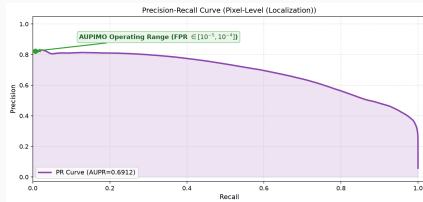
- AUROC relies on False Positive Rate (FPR):
  - With  $\approx 98.5\%$  normal pixels, TN dominates  $\Rightarrow$  FPR barely moves.
  - SOTA scores all exceed 99%, making differences indistinguishable.

## Metric Solution

- **AUPIMO:** Integrates localization over the operational ultra-low FPR window  $[10^{-5}, 10^{-4}]$  on normal images.
- **Dual Thresholds:** We calibrate distinct *image-level* and *pixel-level* anomaly thresholds.
- **F1-Score / PR Curve:** Directly measures defect capture vs false alarms without TN dilution.



Threshold tradeoff curve & AUPIMO range



Precision-Recall curve with marked AUPIMO operating range

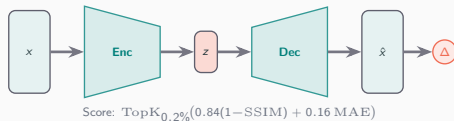
# Models

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# Model Selection & Architecture Overview

## 1. Convolutional-Autoencoders

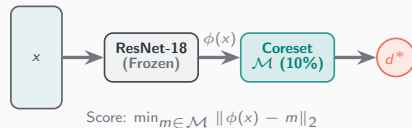
Paradigm: Generative Reconstruction · Built from scratch



- **Complete control:** Full freedom over latent bottleneck dimensions, skip connections, and loss constraints.
- **Intuition:** Trained on normal images; unable to reconstruct defects.
- **Trade-offs:** Prone to blurriness or identity shortcuts; sensitive to reconstruction thresholds.

## 2. PatchCore + ResNet-18 (Anomalib)

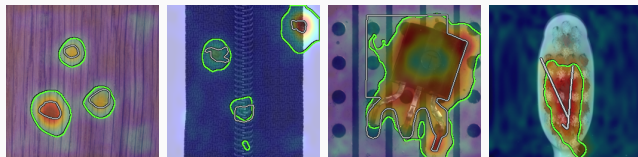
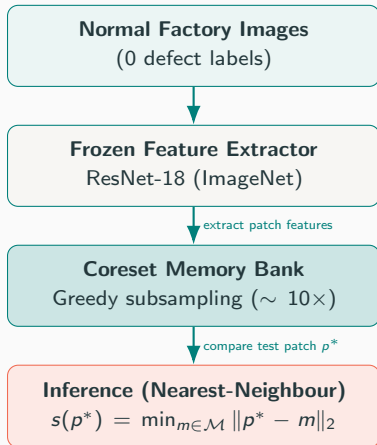
Paradigm: Deep Feature Memory Bank · High accessibility



- **High accessibility:** Zero model training required; out-of-the-box integration via Intel's `anomalib`.
- **ResNet-18 backbone:** Compact ImageNet representations; mid-level layers capture both texture and shape.
- **Coreset sampling:** Compresses the memory bank by 90%+ without losing local anomaly detection accuracy.

# PatchCore: Transfer Learning & Architecture

## Transfer Learning Paradigm



Wood, zipper, transistor, toothbrush. White = GT; green = predicted mask (pixel  $q_{0.99}$ ).

## Key Features

- **Cold-start eliminated:** Deployment without defect collection.
- **IP Protection:** Memory bank contains embeddings, not raw model weights.

## Why it excels

- Mid-level layers capture local textures perfectly.
- No catastrophic forgetting (frozen backbone).
- ResNet-18 guarantees predictable GPU memory bounds.



# Keras CAE – Autoencoder Learning Process

## The Concept

### Training Phase:

- Trained *excl.* on defect-free images.
- Learns the structural grammar of "normal" surfaces.

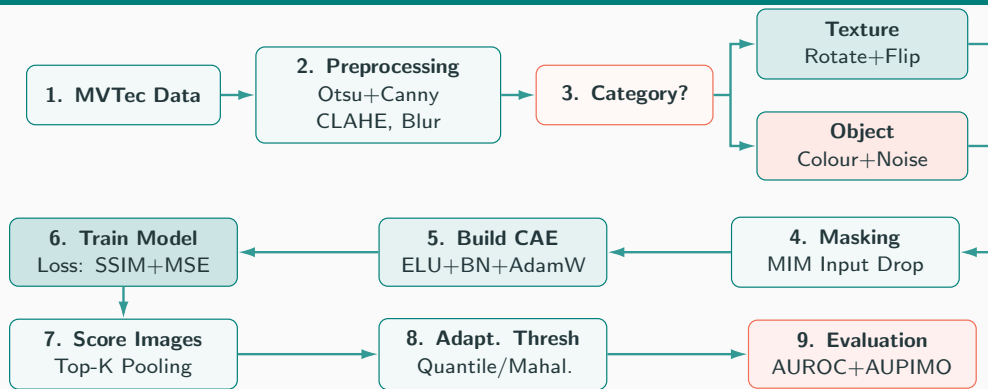
### Inference Phase:

- On defective images, anomalies are reconstructed as "normal" parts.
- Reconstruction fails at the defect sites  $\Rightarrow$  high reconstruction error.
- High reconstruction error  $\Rightarrow$  Anomaly Signal.

## Key Design Decisions

- **Masked Image Modelling (MIM):**  
25% of patches randomly masked before encoding. Forces the model to understand global context, prevents "identity mapping".
- **SSIM + MSE Loss:**  
 $\mathcal{L} = \alpha (1 - \text{SSIM}) + (1 - \alpha) \text{MSE}$ .  
MSE captures global variance, while SSIM enforces high structural and perceptual fidelity.
- **Architectural Choices:**  
**ELU** no dying neurons + zero-centered mean.  
**BatchNorm** stabilizes covariate shifts.  
**AdamW** enables decoupled weight decay.

# Keras CAE – End-to-End Pipeline



## Automated Pipeline & Optuna Integration

The pipeline is dynamically orchestrated. Hyperparameters (Preprocessing/Model) (e.g., augmentation strategy, latent dimensions) optimized using Optuna targeting **Pixel AUPIMO**.

# Evaluation Protocol & Benchmarks

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# Zero-Leakage Evaluation Protocol

## The Core Principle

Every evaluation result in this presentation is produced under a deterministic, audit-compliant split that keeps the test set completely unseen until final scoring  $\Rightarrow$  **zero-test-leakage**.

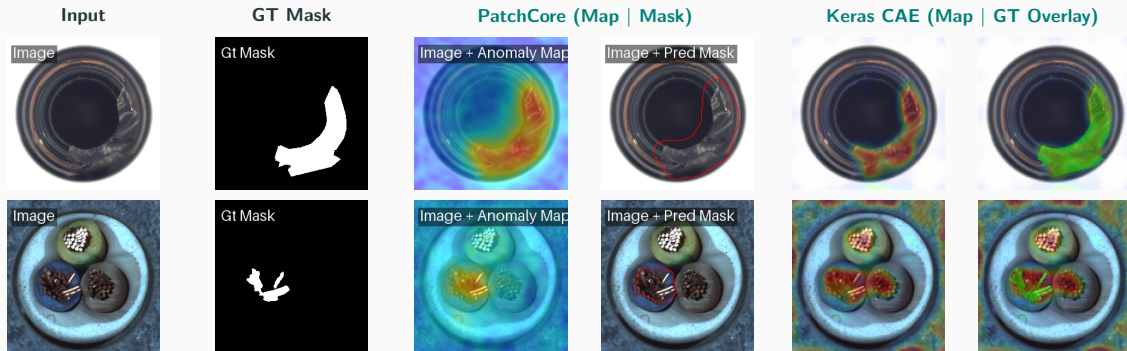
## Peak Performance (PatchCore vs CAE)

| Category        | Metric   | PatchCore    | Keras CAE    |
|-----------------|----------|--------------|--------------|
| Bottle (Object) | Image F1 | 0.962        | <b>0.992</b> |
|                 | AUPIMO   | <b>0.982</b> | 0.507        |
| Macro Average   | Image F1 | <b>0.929</b> | 0.495        |
|                 | AUPIMO   | <b>0.603</b> | 0.058        |

## Canonical Evaluation Settings

- **Split Size:** 85% fitting / 15% validation
- **Test Set:** held out - normal + anomalies
- **Resolution:**  $256 \times 256$  pixels.
  - Bilinear interpolation for continuous heatmaps.
  - Nearest-neighbor + binarization for GT masks.
- **Threshold:** 95th percentile ( $\alpha = 0.05$  false-alarm budget).
- **AUPIMO:** Standard bounds  
 $\text{FPR} \in [10^{-5}, 10^{-4}]$ , 50k thresholds.

# Qualitative Results - Localisation Comparison



## Key Observations:

- **PatchCore:** Heatmaps sharply bound the defect, leading to precise predicted masks.
- **Keras CAE:** Reconstruction errors diffuse. While broadly highlighting anomalous regions, they fail at fine-grained pixel localization.

# Business Interpretation

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# Economic Error Asymmetry — The Cost Model

## Two types of error, very different costs

**False Alarm** (FP): good part rejected.

Cost: line stoppage, lost throughput, scrapped unit

e.g. *Screw*:  $c_{FP} \gg c_{FN}$

**Missed Defect** (FN): bad part ships

Cost: recall, liability, safety incident

e.g. *Pill*:  $c_{FN} \gg c_{FP}$

**Optimal threshold**  $t^* = \frac{c_{FP}}{c_{FP} + c_{FN}}$

High  $c_{FN}$  (pills) pushes  $t^*$  toward zero - catch everything. High  $c_{FP}$  (screws) pushes  $t^*$  toward one - avoid false alarms.

| Category            | Obj.      | $t^*$                      |
|---------------------|-----------|----------------------------|
| Fasteners (Screw)   | $F_{0.5}$ | $\approx 0.9999$           |
| Packaging (Bottle)  | $F_1$     | $\approx 0.0003$           |
| Automotive (M. Nut) | $F_1$     | $\approx 4 \times 10^{-6}$ |
| Life Sci. (Pill)    | $F_2$     | $\approx 10^{-8}$          |

Cost profiles are illustrative estimates from composite industry case studies (for the sake of the project - guesses). They demonstrate the spectrum of liability, not precise financial figures.

# Threshold Calibration & Operating Priorities

## Normal-validation calibration

$$\tau_{\alpha} = \text{Quantile}_{1-\alpha}(\text{AnomalyScores}_{\text{normal, val}})$$

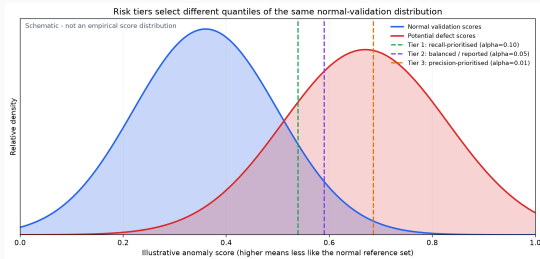
$\alpha$  is a proposed false-alarm budget on held-out normal examples, not a guaranteed production rate.

## Proposed priorities

**Recall priority** ( $\alpha \approx 0.10$ ): lower gate, more review.

**Balanced starting point** ( $\alpha = 0.05$ ): reported benchmark setting.

**Throughput priority** ( $\alpha \approx 0.01$ ): higher gate, fewer flags on normal validation data.



Schematic only. Defect recall and real-line false alarms must be checked with plant data for each proposed setting.

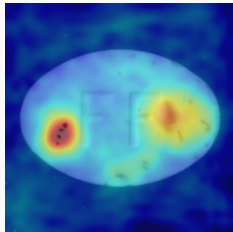


# Operator Evidence for Review

**A flagged image needs visible evidence and a human decision path.**



Input image



Anomaly map



Predicted contour

PatchCore highlights suspected defect regions; the heatmap does not establish the cause.

Borderline flags should go to quality engineers for review during a controlled pilot. Review time and operator agreement still need measurement.

# Conclusion & Next Steps

## What We Demonstrated

- EDA revealed the limitations of conventional evaluation metrics.
- Developed and compared convolutional autoencoders and PatchCore.
- Established a consistent evaluation and threshold-calibration framework.

## What Comes Next

- Validate on representative production data.
- Calibrate thresholds against plant-specific risks.
- Run a controlled industrial pilot before deployment.

From understanding defects to building a model, and from model performance to industrial decisions.

**Thank you**

Questions & Discussion

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